

PAPER_475 — UQFF Sub-Term Physics Modules Catalogue (44 Standalone Calculators)

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Abstract

This whitepaper catalogues 44 standalone sub-term physics calculator modules that constitute the atomic building blocks of the UQFF and MUGE frameworks. Each module encapsulates a single physical quantity — from aether vacuum densities and buoyancy coupling constants to solar cycle frequencies and stellar surface temperatures — as a self-contained C++ class. This catalogue provides a reference index for module equations, key parameter values, output ranges, and integration targets within MAIN_1_CoAnQi.cpp and CondensedPhysics.py.

1. Introduction

The 44 sub-term modules were extracted from grok_share_b0a3dc1d.txt (10,420 lines) and represent the granular physics underpinning the UQFF unified field equation:

$$F_{U,Bi,i} = \int [\text{Ug coupling} \cdot \text{vacuum terms} \cdot \text{buoyancy} \cdot \text{aether} \cdot \text{stellar params}] dx$$

Each module is implemented as a C++ class in modules/subterms/ with standard interface: computeX(), updateVariable(), getEquationText(), printVariables().

2. Module Catalogue

Category A: Vacuum Energy Modules

#	Module	Key Equation	Key Value
1	AetherVacuumDensityModule	$E_A = E_A/c^2$	$7.09 \times 10^{-3} \text{ J/m}^3$
2	UniversalInertiaVacuumModule	E_A [energy]	$7.09 \times 10^{-3} \text{ J/m}^3$
3	ScmVacuumDensityModule	$\text{vac_SCm} = _UA/10$	$7.09 \times 10^{-3} \text{ J/m}^3$
4	UaVacuumDensityModule	$UA = E_UA/c^2$ [mass]	$7.88 \times 10^{-3} \text{ kg/m}^3$
5	BackgroundAetherModule	$E_A = (_A/c^2) _$	$_A = 7.09 \times 10^{-3} \text{ J/m}^3$

Category B: Coupling Constants

#	Module	Key Equation	Key Value
6	AetherCouplingModule	$A_ = g_ + T_s^{\wedge}$	$1/E_s_total \quad 1e-15$

#	Module	Key Equation	Key Value
7	UgCouplingModule	k_i weights Ug_i	$k=1.5, k=1.2, k=1.8, k=1.0$
8	BuoyancyCouplingModule	$F_{bi} = - \rho_i U_{gi} \Omega_g \dots$	$\rho_i = 0.6$ uniform
9	InertiaCouplingModule	$I_c = M r^2 \Omega^2 / c^2$	Sun: 7.8×10^{-1}
10	UgIndexModule	$Ug_{\{i,n\}} = G M / r^2 Q_{i,n}$	4×26 array

Category C: Solar/Stellar Parameters

#	Module	Key Equation	Key Value
11	SolarWindBuoyancyModule	$v_{sw} = 0.001 \times (1 + A \sin(t))$	$v_{sw} = 0.001$ baseline
12	SolarWindModulationModule	$v_{sw}(t) = v_0(1 + A \sin(t))$	$v_0=400$ km/s, $A=0.2$
13	SolarWindVelocityModule	$v_{sw} [400,800]$ km/s	$v_{fast} = 750$ km/s
14	SolarCycleFrequencyModule	$\tau_{sc} = 1/(11 \text{ yr})$	2.88×10^{-8} Hz
15	HeliosphereThicknessModule	$R_{HP} - r_{TS}$	~ 30 AU ($\sim 4.5 \times 10^{12}$ m)
16	StellarMassModule	$M_s(t) = M_0(1 - \alpha_{ML} t)$	$\alpha_{ML} = 1.4 \times 10^{-1} \text{ s}^{-1}$
17	StellarRotationModule	$\omega_s = 2 / P_{rot}$	Sun: 2.87×10^{-8} rad/s
18	SurfaceMagneticFieldModule	$B_{mag} = M_{mag} / 4 r^3$	Magnetar: 4.4×10^{13} T
19	SurfaceTemperatureModule	$T_s = (L / 4 r^2)^{0.25}$	Sun: 5778 K
20	MagneticMomentModule	$\mu = I A_{vort}$	Solar: $\sim 1 \times 10^{21} \text{ A} \cdot \text{m}^2$

Category D: Galactic/Astrophysical Parameters

#	Module	Key Equation	Key Value
21	GalacticDistanceModule	$r_{leg} = \text{virial radius}$	MW: 2.55×10^2 m
22	GalacticSpinModule	$\Omega_g = 2 / T_{gal}$	MW: 7.3×10^{-1} rad/s
23	GalacticBlackHoleModule	$M_{BH} (M_\odot)$	Sag A*: 8×10^3 kg
24	FeedbackFactorModule	$f_{env} = f_{AGN} + f_{SN} + f_{SF}$	$f_{AGN}=0.1, f_{SN}=0.05$
25	MagneticStringModule	$F_s = (I^2 / 4) \ln(L/a)$	$\sim 10^2$ N (cosmic string)
26	Ug3DiskVectorModule	$Ug3_{disk} = G M_{disk} / r^2 (h/r)$	MW: $h/r = 0.07$
27	Ug1DefectModule	$Ug1_{corr} = Ug1(1 - \alpha_{def})$	$\alpha_{def} = 0.05-0.15$

Category E: Quantum & Field Calculators

#	Module	Key Equation	Key Value
28	DPMModule	26-sphere: $(x-h)^2 + \dots = r^2$	$SCm E = 10^{-2} \text{ J}$
29	UnifiedFieldModule	$F_U = \sum k_i Ug_i + F_{bi} \dots$	Orchestrator
30	StressEnergyTensorModule	$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu}$	Trace = $-\rho + 3p$
31	QuasiLongitudinalModule	$Q_L = E^2 / 2$	$\sim 1e-12 \text{ J/m}^3$
32	OuterFieldBubbleModule	$\phi = r \exp(H t)$	At 10 Gyr: $r = 1.28 \text{ r}$
33	ReciprocatationDecayModule	$\phi \exp(-t / \tau_{rec})$	$\tau_{rec} \sim 1 \text{ Gyr}$
34	ScmPenetrationModule	$\alpha_{SCm} = (\alpha / \alpha_{SCm})^{0.5}$	London-analog depth
35	ScmReactivityDecayModule	$d[SCm]/dt = -k_r [SCm]$	$[SSq]=0.57, k_r \sim 1e-18 \text{ s}^{-1}$

#	Module	Key Equation	Key Value
36	ScmVelocityModule	$v_SCm = c/n_SCm$	$n_SCm = 1.0000001$
37	PiConstantModule	$= 4\Sigma(-1)^k/(2k+1)$	Leibniz series
38	CorePenetrationModule	$= (_{core}/_{avg})^n r_{core}$	NS core: $\rightarrow 0$
39	NegativeTimeModule	$g(t<0) = g \cos(t)$	$f_TRZ = 0.1$
40	TimeReversalZoneModule	$f_TRZ = r_TRZ/r$	Canonical: 0.1
41	HeavisideFractionModule	$H(f): 0 (f<0), 1 (f>0)$	Threshold: 0
42	StepFunctionModule	(x) : boxcar windows	Regime switching

Category F: Index / Utility Modules

#	Module	Key Equation	Key Value
43	GalacticBlackHoleModule	$\phi_{BH} = G M_{BH}/r^2$	At $r=5.5 \times 10^1$ m (Sag A*)
44	InertiaCouplingModule	$I = 1 + I_c$	Ug3 relativistic boost

3. Integration Targets

3.1 MAIN_1_CoAnQi.cpp

Sub-term modules are candidates for batch extraction as `PhysicsTerm` subclasses:

```
// Example: ScmVacuumDensityModule → ScmVacuumDensityTerm
class ScmVacuumDensityTerm : public PhysicsTerm {
public:
    double compute(const SystemParams& params) override {
        return 7.09e-37; // _vac_SCm [J/m³]
    }
    std::string getName() const override { return "SCm Vacuum Density"; }
};
```

Target integration point: **Batch 24** in SOURCE1-116 registry near line 6688+.

3.2 CondensedPhysics.py

Each sub-term module maps to a `Calculator` class method:

```
class SubTermCalculator:
    def compute_aether_vacuum_density(self) -> dict:
        # Returns _vac_A = 7.09e-36 J/m³ with equation text
```

Target: new `UQFFSubTermCalculator` in `CondensedPhysics.py` Part 5 (after line 60000).

3.3 index.js

Module constants exportable as:

```
const SUB_TERMS = {
  rho_vac_UA: 7.09e-36,    // J/m3
  rho_vac_SCm: 7.09e-37,  // J/m3
  beta_i: 0.6,             // buoyancy coupling
  kappa: 0.0005/86400,    // per-second DPM rate
  SSq: 0.57,              // SCm calibration
};
```

4. Calibration Constants Summary

Constant	Value	Source Module
	0.0005/day	DPMModule
[SSq]	0.57	ScmReactivityDecayModule
_i	0.6 (all i)	BuoyancyCouplingModule
_sw	0.001	SolarWindBuoyancyModule
	$\sim 10^{-1}$	AetherCouplingModule
H_SCm	0.99	ScmVacuumDensityModule
k _{,k,k,k}	1.5/1.2/1.8/1.0	UgCouplingModule
f_TRZ	0.1	TimeReversalZoneModule

5. Scientific Context

The 44 sub-term modules collectively represent a field-theoretic decomposition of gravity that accounts for:

- **Vacuum energy hierarchy:** $_UA : _SCm : _cosm = 10 : 1 : 0.001$
 - **Coupling hierarchy:** $k_{\text{(string/rotation)}} > k_{\text{(dipole)}} > k_{\text{(charge)}} > k_{\text{(vacuum)}}$
 - **Time evolution:** DPM birth model $\rightarrow$ inflation $\rightarrow$ SCm decay $\rightarrow$ solar wind modulation $\rightarrow$ present
 - **Multi-scale validity:** Sub-parsec (heliosphere) to Gpc (galaxy clusters)
-

6. Conclusion

This catalogue documents 44 atomic physics calculator modules that implement the sub-term physics of the UQFF/MUGE framework. The modules span vacuum energy, coupling constants, stellar parameters, galactic scales, and quantum field calculators. They constitute the `modules/subterms/` library and are ready for integration as `PhysicsTerm` subclasses in `MAIN_1_CoAnQi.cpp` or as Python calculator methods in `CondensedPhysics.py`.

Source: `grok_share_b0a3dc1d.txt` L1502–9356 (44 class definitions)

Header index: `modules/subterms/sub_terms_index.h`

Tags: sub-terms, catalogue, vacuum-density, coupling-constants, solar-parameters, UQFF, MUGE, integration